COMP 486: Assignment 3 Game Design Document

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Assignment 3 will be a continuation of my HTML game from Assignment 2. I will briefly outline some of the repeat items from Assignment 2 below before discussing the new items I will be adding to the game in Assignment 3.

Game Title: Sudsy's Cleaning Frenzy - A Survival Game
Backstory:

- Sudsy Spongerton (a sponge) needs to ensure the entire kitchen area remains clean as their enemies (vials of messy liquids) try to continually deter them and spoil their cleaning plans.

Characters:

- I am going to build all character sprites myself using simple image editors and shapes.
- Main Character:
    - Sudsy Spongerton
- Enemies (All shaped as different coloured glass jars/ vials with legs):
    - A small enemy (Green grease)
        - One shot to destroy
    - A small fast enemy (different coloured small enemy)
        - One shot to destroy
    - A larger enemy (Yellow grease)
        - 2 shots to destroy
    - A large 'invincible' enemy (larger version of same character)
        - Can walk through walls
        - More shots to destroy.

Gameplay:
Gameplay Mechanics

- Sudsy Spongerton:
    - MOVEMENT: controlled with the arrow keys on the keyboard.
        - It will be stop and go movement, where when not pressing, sudsy will stop.
    - SHOOTING: shoots soap sponges by pressing spacebar.
    - LIVES: Has life out of 100%.
    - PICKUPS/UPGRADES:
        - Life: adds a life to Sudsy, +20%.
        - Bubble ammo: Sudsy has limited ammo, so will need to reload.
            - This will randomly spawn in the map after a set amount of time.
        - WEAPONS:
            - Purchased by detecting overlap, and pressing a button when overlapping. Uses points to buy these.

            - Soap bombs: Purchased, when shot, sit where Sudsy was, destroy 3 enemies when collided with.
            - Bouncing bullets: Purchased, when shot, destroy all enemies they collide with, disappear after bouncing 3 times off of any wall (detect wall collisions).
            - Rapid Fire ammo: Purchased. Will use the same bullets as regular ammo, but faster firing rate.
            - Ammo limit increase: purchaseable, which increases the number of bullets each item can carry.
- GATES:
    - Run into the gate to see the cost of it. Collision is detected.
    - If Sudsy has enough money to buy the gate, it opens. This makes gates something to avoid when you have a lot of money too.
    - If sudsy cannot afford it, the cost displays.
    - Gate costs will increase with each gate that is opened:
        - Cost = 100+(numGatesUnlocked+1)*60
        - As per logic provided with A2. This makes gates affordable but not too cheap for the user based on the points they are making.
    - Enemies can walk through the gates.
- Enemies:
    - Chase the main character directly
    - The large enemy that can walk through walls is added so that the user cannot just sit in a hidden spot, given the enemies are not "smart".
    - When enemies collide with Sudsy, sudsy loses 0.02% of their health.
    - Enemies will collide with bullets and weapons.
        - If they have been damaged enough, they disappear.
        - If they take multiple shots to kill, there will be some feeedback to the user showing the enemy is damaged.
- GUI:
    - The game is top down view, the same as Assignment 2.
    - The viewport is focussed on the main character.
- HUD:
    - The top HUD will show health and weapons.
    - The bottom of the screen will show points and level value.
    - The ammo limits will be display in the top HUD if room or visible in the pause screen.

Game Design:

- The game is developed in Phaser V3.
- Maps will be created using TILED and imported into Phaser as JSON files.
- Collision and overlap handling will be handled by Phaser's Internal Physics engine.
- Enemy Spawning:

        - For more levels enemies will spawn from the side, much like Assignment 2, however, for one level, I plan to have enemies spawn from the middle of the screen.
        - This will be on a timed loop (using `this.scene.time.addEvent()`)
            - As was done in Assignment 2
        - On every second wave of enemies, one of the small enemies will be made fast, and the large enemy will be made the largest enemy that can walk through walls.

Game Flow:

- The user will be greeted by the main menu, where they select their chosen level.
    - I plan to make 3 levels, all with different features to appeal to various users and gameplay. (small level, medium sized level, large level in size).
        - The smallest level will be more intense and close quarters.
        - The medium level will be more exploratory.
        - The largest level will be open and simply about long term surviving and running around.
        - All maps are predetermined, finite, and built ahead of time.
- Once they select their level, they will see the title page for that level, where they can see the level name, map, and even view high scores for that level.
- Once they click start, they game will be taken to the paused screen, where, when ready, the user can click start and begin playing.
- The player can click to pause the screen.
- They destroy enemies to get points, and use points to open gates and buy weapons.
    - Since there are 4 weapons, I want to use a priority system for weapon usage.
        - So the weapons are always used in a certain order, but only if the weapon has ammo.
        - I am thinking Bomb > bouncing bullet > Rapid Fire> regular ammo.
        - As the bombs would be something you want to place strategically maybe before the next level. And the rest is used when the level begins.
- Once all enemies of a level are destroyed, the next level begins immediately.
- Once the player's health hits 0%, the game over screen appears, where they can enter their highscore (stored locally using LocalStorage) or simply return to the main menu.

Game Rules:

- Survive to the highest level.
- Destroy all enemies to get to the next level.
    - Destroying enemies gets the user points.
- Unlock rooms to have a larger playing area and find weapons to purchase.
- Points are used to purchase gate opening and weapons.
- Weapons have limited ammunition that they can hold at one time.

Victory Conditions:

- Play until the user's health hits 0%.
- High Score = The level the user reached.

NEW ADDITIONS FROM ASSIGNMENT 2:

The new items I plan to add/ have added for Assignment 3 are as follows:

1. Restructured code to have multiple scenes:
    a. For main menu and Map scenes
    b. Allows for reuse of the actual game scene
2. Main menu
    a. This can be used to:
        i. Select the desired map (3 maps in total)
            1. Click on map and can:
                a. hit play
                b. See high scores for that map
                i. Goes to a high scores scene.
                c. Or return to the main menu.
    b. The main-menu can be reached from game over and pause screens.
3. Pause scene
    a. Replace the "click" to pause with a press of "P".
4. Restart level
    a. Restart buttons to be added to the pause and game over scenes.
    b. Also press "r" during gameplay to restart.
5. Stop Sudsy from being pushed by enemies.
    a. Sudsy will not be pushed into the walls or gates anymore.
    b. Allows sudsy to run through enemies, while receiving damage.
6. Add sounds to gameplay.
    a. Sound for shooting
    b. Sound for Sudsy+enemy collision
    c. Sounds for gate collisions and unlocking
    d. Sounds for ammo maxed out or cannot afford an item
7. Make gates and pickups draw as an object layer in Tiled and not manually place using coordinates.
8. Game Over Screen:
    a. Can access the main menu.
    b. Can restart level.
    c. Allows user to enter their name →
        i. this (and their score) are added to the high scores table
            1. Stored locally with LocalStorage
            2. Each level will have its own high score associated.
9. Add a game store:
    a. This is unlocked through the regular gates, it will look like a room (or many rooms), the user can walk around and purchase:
        i. weapon(s)
            1. A soap bomb that can be kicked at enemies or left out and explodes when enemies hit it.
            2. Gun which shoots faster.

        3. Bouncing bullets (make them larger and easier to see bouncing around the screen).
        4. Upgrade Ammo
            a. Doubles the amount of ammo you can hold.

10. New levels (kitchen, drain pipes, bathroom):
    a. 1. KITCHEN Counter (original level)
        i. This will be upgraded to have a store.
        ii. Keep this level a bit more close quarters.
    b. 2. PIPES: I think an exploratory level, if someone wants to explore a bit more. Could look like pipes connected. Somewhat maze-like.
    c. 3. BATHROOM: enemies emerge from the middle, but the rest of the map will be open. Allows the user immediate access to all weapons.
11. Enemies:
    a. Spawning logic:
        i. Spread out a bit more, change some areas of the map.
    b. Some fashion of seeing enemy damage (either a health bar or changing their tint).
    c. A large "boss enemy" every second wave to flush the user out of hiding anywhere.
    d. A faster enemy in every second wave.
12. Gate unlock:
    a. Left similar, but just more gates as they also add another obstacle to avoid when saving up money.

COMP 486: Assignment 3 Test Plan

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Test Item	Input		Expected Output	Actual Output	Results/Remarks
Controls	Sudsy movement		Arrow keys should make sudsy move in all directions. Combination key presses should be summative.	Arrow keys allow sudsy to move in all directions, even diagonals. Summative key presses work as expected.	Complete
	Sudsy Shooting	Regular Ammo	The space bar should make sudsy shoot a spongey bullet.	The space bar makes sudsy shoot a bullet.	Complete
		Rapid Fire	When sudsy has Rapid ammo, the spacebar should shoot this before regular ammo.	The spacebar makes rapid ammo first, followed by regular ammo.	Complete.
		Bouncing Bullet	When Sudsy has Bouncing ammo, it is shot before rapid ammo.	The spacebar makes bouncing ammo shot before rapid ammo.	Complete.
		Soap Bombs	When sudsy has soap bombs, they are used before bouncing ammo.	The spacebar shoots bombs before bouncing ammo.	Complete.
Display (various screens and animations)	Main Menu Scene		The main menu should display 5 images above the game title and 3 buttons for each level.	The main menu displays the images, the title, and the three buttons.	Complete
	Map Info Scene		The map info scene should display the map name, description,	This page displays all expected traits for all three maps correctly.	Complete.

		image, and 3 buttons (start, highscores, and main menu).		
	HighScore Scene	This is expected to display the top 10 scores of the level in order of levels reached. If no scores available, it should say this.	With no scores, it says no score available. Otherwise the scores are listen in order of highest to lowest level reached, and only displays the top 10.	COMPLETE
	Pause Scene	The pause scene should appear when the "p" is pressed during game play and when the level first begins.	Pause scene appears in both circumstances with the correct wording.	COMPLETE
		The pause scene should display the ammo limits of each weapon, which should update when the ammunition is upgraded.	The ammunition limits are displayed and updated when purchased.	Complete.
	Restart Level	When the user presses 'r' the game should restart on the current level, and all data reset.	When the user presses 'r', all data resets and they are able to continue gameplay on a new set.	COMPLETE
	Game HUD	The game HUD should display the players health and ammunition amounts.	The HUD displays the health and ammunition amounts with no overlaps.	Complete.

		The game HUD should display the player's points total and current level in the bottom of the screen.	These values are displayed correctly in all levels.	Complete.
	Sudsy	Sudsy should stand still when not moving, but appear to be running/ driving when pressing the arrow keys.	Sudsy's spritesheet animations give the illusion of Sudsy running/ driving in all directions.	Complete.
	Small Enemy	The small enemy should always appear to be walking in all directions.	The small enemy walks and uses all views of the spritesheet.	Complete
	Large Enemy	The large enemy should always appear to be walking in all directions.	The large enemy walks and uses all views of the spritesheet.	Complete
	Largest Enemy	The largest enemy should always appear to be walking in all directions.	This enemy walks in all directions and follows the user.	Complete
	Health Pickup	The health pickup should appear to be flashing 3 different colours.	The upgrade flashes 3 different colours in a slow fashion.	Complete
	Ammo Pickup	The pickup should flash 3 different colours to draw the viewers attention.	The pickup flashes 3 different colours continuously.	Complete.
	Bullet	When a bullet is fired, it should appear as a small sponge	The bullet appears as a small sponge.	Complete.

			going across the screen.		
	Soap Bomb	Pickup Display	The pickup for this item should appear where it was located in the Tiled JSON Map.	This item appears where located in the Tiled map display.	Complete. Formatting needs to consider Tiled uses top left, Phaser uses center (setOrigin()).
		Shot	When shot, the item should appear where Sudsy is located and remain in that position.	When shot, the item is where Sudsy is and remains in that position.	
	Bouncing Bullet	Pickup Display	The pickup for this item should appear where it was located in the Tiled JSON Map.	This item appears where located in the Tiled map display.	Complete. Formatting needs to consider Tiled uses top left, Phaser uses center (setOrigin()).
		Shot	When shot the item should appear and fire in the direction sudsy is facing.	The object fires in the direction sudsy is facing.	Complete.
	Rapid Fire	Pickup Display	The pickup for this item should appear where it was located in the Tiled JSON Map.	This item appears where located in the Tiled map display.	Complete. Formatting needs to consider Tiled uses top left, Phaser uses center (setOrigin()).
		Shot	When a bullet is fired, it should appear as a small sponge going across the screen in front of sudsy.	The bullet appears as a small sponge in front of Sudsy.	Complete.

	Ammo Limit Upgrade		The pickup for this item should appear where it was located in the Tiled JSON Map.	This item appears where located in the Tiled map display.	Complete. Formatting needs to consider Tiled uses top left, Phaser uses center (setOrigin()).
Collisions (+HUD Updates)	Sudsy	Wall	Sudsy will not be able to go through the blue walls when colliding.	Sudsy cannot go through the blue walls, collision is detected.	FIXED. Sudsy can no longer be pushed through the wall by enemies.
		Small Enemy	Sudsy should flash to show the contact, and the health should reduce.	This works as expected, there was an error where enemy would fully overlap and health does not decrease. This seems to be fixed with setting Sudsy's to non-pushable.	Small error appears to be resolved. Does not impede gameplay.
		Large Enemy	Sudsy should flash to show the contact, and the health should decrease.	This works as expected, there was an error where enemy would fully overlap and health does not decrease. This seems to be fixed with setting Sudsy's to non-pushable.	Small error appears to be resolved. Does not impede gameplay.
		Largest Enemy	Sudsy should flash to show the contact, and the health should decrease.	This works as expected, there was an error where enemy would fully overlap and health does not decrease. This seems to be	Small error appears to be resolved. Does not impede gameplay.

				fixed with setting Sudsy's to non-pushable.	
		Life Pickup	The health upgrade should disappear, and the HUD should increase the health up to a max of 100%.	The upgrade disappears and the HUD is updated and does not exceed 100%.	Complete.
		Regular Ammo Pickup	The upgrade should disappear and the HUD ammo should increase but not exceed the limit.	The upgrade disappears, the HUD increases to a max of the limit, and this also matches the number of bullets Sudsy can fire.	Complete.
		Soap Bomb Pickup	When sudsy overlaps, a message displaying the cost of the item should display. This should disappear when sudsy leaves. If the user hits A when overlapping, they will either purchase the item (if enough money) or be indicated they cannot afford it (discussed in Additional User Feedback below).	The text appears when sudsy overlaps, disappears when sudsy leaves. Pressing A lets user buy the item or indicates clearly that the user cannot afford it when this is the case.	Complete. Text had a stroke added to make clearer to read.
		Bouncing Bullet Pickup	When sudsy overlaps, a message displaying the	The text appears when sudsy overlaps, disappears	Complete. Text had a stroke added to make clearer to read.

			cost of the item should display. This should disappear when sudsy leaves. If the user hits A when overlapping, they will either purchase the item (if enough money) or be indicated they cannot afford it (discussed in Additional User Feedback below).	when sudsy leaves. Pressing A lets user buy the item or indicates clearly that the user cannot afford it.	
		Rapid Fire Pickup	When sudsy overlaps, a message displaying the cost of the item should display. This should disappear when sudsy leaves. If the user hits A when overlapping, they will either purchase the item (if enough money) or be indicated they cannot afford it (discussed in Additional User Feedback below).	The text appears when sudsy overlaps, disappears when sudsy leaves. Pressing A lets user buy the item or indicates clearly that the user cannot afford it.	Complete. Text had a stroke added to make clearer to read.
		Ammo Upgrade Pickup	When sudsy overlaps, a message displaying the cost of the item should display.	The text appears when sudsy overlaps, disappears when sudsy leaves.	Complete. Text had a stroke added to make clearer to read.

			This should disappear when sudsy leaves. If the user hits A when overlapping, they will either purchase the item (if enough money) or be indicated they cannot afford it (discussed in Additional User Feedback below).	Pressing A lets user buy the item or indicates clearly that the user cannot afford it.	
		Gate (too expensive)	When sudsy bumps into the door, it should display the cost to clean it up.	The cost is displayed as expected.	FIXED. Enemy can no longer push Sudsy through the gate.
		Gate (can afford)	If Sudsy bumps into the slime and has enough points to clean it, the wall cleans after a set time and then disappears.	Works as expected, cleans and then disappears, and the points are deducted from sudsy. Each successive cleaning costs more. The costs are consistent with the formula chosen.	Complete.
	Bullet (regular and rapid fire)	Wall	When a bullet collides with the wall it disappears.	The bullet collides with the wall and disappears.	Complete.
		Small Enemy	If bullet collides with small enemy, enemy and bullet disappear, points total should increase	Bullet and enemy disappear on collision, points total increments appropriately.	Complete.

			by 10.		
		Large Enemy	If bullet collides with large enemy, bullet disappears, enemy disappears only after hit twice, then points increment by 20.	Bullets and enemy disappear after 2 bullet collisions, points total increments appropriately.	Complete.
		Largest ("BOSS") Enemy	If bullet collides with large enemy, bullet disappears, enemy disappears only after hit twice, then points increment by 50.	Bullets and enemy disappear after 5 bullet collisions, points total increments appropriately.	Complete.
	Soap Bomb (shot)	Small Enemy	As soon as enemy collides, enemy should disappear. Soap bomb should disappear after 3 enemy collisions.	Small enemy disappears, bomb disappears after 3 enemies collide.	Complete.
		Large Enemy	When enemy collides, changes colour, and will disappear after 2 hits. Bomb disappears after a total of 3 hits.	The bomb makes enemy lose life, and can tolerate one more hit after wards.	Variability noted. Feature works as created and is suitable for gameplay.
		Largest ("boss") Enemy	The largest enemy should slowly decrement from being hit by the bomb, the bomb will be taken out by one large enemy.	This works but since the large enemy uses a timer to limit collisions, it does depend on if it hits bomb dead on, or glances past it	Variability noted. Feature works as created and is suitable for gameplay.

				or walks vert/ horiz through it. The more time touching = the more time colliding with the bomb and the more damage received. Bomb disappears if it takes out the large enemy.	
	Bouncing Bullet (shot)	Wall	The bullet should bounce off of the wall, for a total of 3 bounces, where on the 4th it disappears.	The bullet bounces a total of 3 times off of the wall.	Complete.
		Small Enemy	When the bullet hits the small enemy, the small enemy is destroyed.	The bullet destroys the small enemy immediately.	Complete.
		Large Enemy Largest "BOSS" Enemy	When the bullet hits the larger enemy, it reduces its life, but does not destroy it on the first impact.	The bullet reduces the health of the larger enemies, but does not destroy them on the first impact. On the second or third impact it destroys them.	Complete.
	Small Enemy	Wall	Small enemy should not be able to walk through the wall. When the enemy is spawned inside of the wall, it should be able to walk out of the wall.	Small enemy cannot walk through the wall, except for the moments when it is spawned inside of the wall. Once in the map, it cannot walk through the wall. One small thing	Resolved Small error. Sudsy can push the enemy into the wall. All walls were designed to be double wide so that the enemy cannot be pushed into another room or outside of the

				is Sudsy can push the enemy into the wall.	map.
		Small Enemy	Small enemies should appear to bump into one another.	Small enemies do bump into one another, but if a lot of them are pushed in the same spot, they do overlap. They then move in unison, so it may look like one, but could be many.	Overlap of enemies noted.
		Large Enemy	Enemies should appear to bump into one another.	Enemies do bump into one another, but if a lot of them are pushed in the same spot, they do overlap. They then move in unison, so they may sometimes appear overlapped.	Overlapping noted.
	Large Enemy	Wall	Large enemy should not be able to walk through the wall. When the enemy is spawned inside of the wall, it should be able to walk out of the wall.	Large enemy cannot walk through the wall, except for the moments when it is spawned inside of the wall. Once in the map, it cannot walk through the wall. One small thing is Sudsy can push the enemy into the wall.	Resolved Small error. Sudsy can push the enemy into the wall. All walls were designed to be double wide so that the enemy cannot be pushed into another room or outside of the map.
		Large Enemy	Enemies should appear to bump into one another.	Enemies do bump into one another, but if several of them	Overlapping noted.

				are pushed in the same spot, they do overlap. They then move in unison, so it may look like one, but could be many.	
Spawning/ Re-spawning	Bullets		When bullets are reused, they should act normally and obey phaser physics.	This was fixed, as need to enable bullets after being disabled.	Complete.
	Small enemies		At the beginning of the level small enemies in waves of 5 should come from the left hand side of the map wall. The waves should be spaced out by 2 seconds.	The waves of small enemies spawn correctly and are cnogruent with the level number determining the number of waves.	Complete.
	Large Enemies		There should be one large enemy in each wave of enemies spawned. Spaced out by the 2 second time interval.	One large enemy is spawned with each successive level. The number of large enemies == the level number.	Complete.
	Largest Enemies		In every second wave, there should be a larger version of an enemy that can walk through walls.	There is one larger enemy that walks through everything to the user in every second wave.	Complete.
	Fast enemies		In every second wave there will be a small enemy tinted all white that moves	There is one fast enemy in every second wave of each level.	Complete.

			faster than every other enemy.		
	Ammo Pickup Respawning		After 20 seconds, the pickup should reappear where it was collected.	The pickup reappears after the time limit! :)	Complete.
	Health Pickup Respawning		After 30 seconds, the health pickup should reappear where it was collected.	The pickup reappears after the set time limit!	Complete
Buttons (on screen)	Main Menu	All	When hovering over these buttons, they should have a tint. This should disappear once the cursor is moved off.	Hover over the text with the cursor adds a tint, this is removed when the cursor moves.	Complete.
		Kitchen	This should load the kitchen map info page.	This loads the kitchen map page.	Complete.
		Pipes	This should load the pipes map page.	This loads the pipes map page.	Complete
		Bathroom	This should load the bathroom map page.	This loads the bathroom map page.	Complete
	Map Info	All	When hovering over these buttons, they should have a tint. This should disappear once the cursor is moved off.	Hover over the text with the cursor adds a tint, this is removed when the cursor moves.	Complete.
		Start	This should take us to the gameScene to start the level.	Works as expected takes to game scene with a fresh game.	Complete.

		Highscores	This should go to the Highscores page described above.	Takes user to highscores for that particular map. This is unique to the map itself.	Complete.
		Return	This should return the user to the main menu.	This takes the user to the main menu. Works for every map.	Complete.
	HighScores	Back	This should return the user to the map info page.	This returns the user to the map info page for each map.	Complete.
	Pause Screen	All	When hovering over these buttons, they should have a tint. This should disappear once the cursor is moved off.	Hover over the text with the cursor adds a tint, this is removed when the cursor moves.	Complete.
		Start/Resume	This should allow the user to continue their game in the same scene with no loss of data.	Data is carried across pauses, and game continues without fault.	Complete.
		Main Menu	This should return the user to the main menu.	This takes the user to the main menu. Works for every map.	Complete.
		Restart	When the user presses restart, the game should restart with all new values.	When pressed, an entirely new game starts.	Complete.
	Game Over Screen	All	When hovering over these buttons, they should have a tint. This should disappear once the cursor is	Hover over the text with the cursor adds a tint, this is removed when the cursor moves.	Complete.

			moved off.		
		Submit	When click submit, you can submit score as name, or BLANK if don't enter anything.	The score is added to the high scores for that level if it is in the top 10. It displays the name OR blank for the entry.	Complete.
		Main Menu	This should return the user to the main menu.	This takes the user to the main menu. Works for every map.	Complete.
		Restart	When the user presses restart, the game should restart with all new values.	When pressed, an entirely new game starts.	Complete.
Additional User Feedback	Cannot afford an item		If a user cannot afford an item the game should: sound alarms, and the POINTS should flash red and enlarge text.	When the user cannot afford (gate, all weapon upgrades) the text flashes and a sound alarms.	Complete.
	Purchase maximum		When the user is trying to buy a weapon but is already maxed out, the weapon in the HUD should flash red.	The amount of ammo flashes red when user is maxed out in ammo for all weapons.	Complete.
	Next level		When the user reaches the next level, a specific sound should play, and the level text should enlarge and flash yellow.	When the user reaches the next level, a specific sound plays, and the level text enlarges and flashes yellow.	Complete.
	Regular Ammo empty		When the ammo is fully empty, it should flash red.	The ammo flashes red when empty and	Complete.

			no other ammo is present.	
	Ammo upgrades flash	When the ammo limits are upgraded, the ammos covered should flash green and the new limits should show in the pause screen.	Ammo covers flash green and new limits are presented.	Complete.
	Enemy burst into bubbles when destroyed.	When enemies destroyed, they should burst into bubbles.	Enemies burst into a bubble when destroyed.	**Removed** after playing as obstructs field of view and is very distracting.
Sounds	Purchase item	When the user purchases any item, a sound should play.	The purchase sound plays when users purchases an item.	Complete.
	Cannot afford	If the user cannot afford an item, a sound should play.	A sound plays when the user cannot afford and weapon or gate.	Complete.
	Gate (cleaning)	When a gate is unlocking (cleaning) a sound should play.	The cleaning sound plays each second of the cleaning.	Complete.
	Shooting	When the user shoots, a sound should play.	A sound plays whenever the user shoots.	Complete.
	Enemy collision	When the user collides with the enemy, a notable sound should play.	When the user collides with any enemy, a sound plays.	Complete.
	Enemy destroyed	When enemy destroyed a sound should	Sound played when enemy destroyed.	**REMOVED**. Constantly going. Was quite

			play.		annoying.
	Increment level		When a user reaches the next level, a sound should play.	The next level sound plays when user reaches the next level.	Complete.
Game Conditions	Game Over		When the user hits 0% life, the game over screen should appear and the user should not be able to do anything beyond return to the menu or submit their score.	When the game ends, the game over screen appears, letting the user submit their score or return to main menu.	Complete.
			If the user submits their score, it only appears if it is in the top 10 for that level.	This works as expected, only the top 10 scores are kept.	Complete.
	Fair Game Conditions	Kitchen	The levels should be challenging, but allow the user to be able to navigate through. The ammo should be suitable, as well as the health upgrades. Items should be affordable, but not too easy to get. The levels should feel harder as we progress.	This level is closer quarters, the first 2 levels are actually tricky, need to time the gate just right. The giant enemies are harder, but it flushes the player out. Level was quite easy without this. Getting ammo take patience, but it is doable. It is easy to get trapped with larger enemy as cannot shoot through walls.	Resolved. This game allows the user to hide out a bit more. This lead to the creation of the larger "BOSS" enemy. Lead to reducing the amount of weapons user can hold.
		Pipes		This level is	Resolved. Map

				slower to begin with, but can get longer into the level. This is probably the least exciting level as the characters do get stuck in it and need to explore around a lot. This does put the player in tricky situations of needing to go and find enemies, which adds a challenge. The goal of exploration was achieved for the level.	layout could be improved for enemy movement, but exploration achieved.
		Bathroom		This level is for the user that wants to run around shooting things, but the game can drag on a little bit, as it is fairly easy to get away from enemies. That is somewhat the point of the level though.	Complete. In all levels, there is a point at which you make almost too much money (>level 16) where the game does get less enjoyable. It is easier to reach this in this level, although that is the point of this level, it could be improved.